The ProbLemma's Channel Season 3 Guide

Season 3 Episode 1: The Index Of Tools

- All the problem-solving approaches studied so far, across seasons 1 and 2, are indexed by the episodes in which they were defined and practiced

Season 3 Episode 2: An Integral Transformation

- Problem S3M1 solved, show that for all θ, x real, with $x > 1$, and $n = 1, 2, 3, 4, 5 \ldots$ it is the case that:

$$\int_0^\pi \left(x + \sqrt{x^2 - 1}\cos(\theta)\right)^n \mathrm{d}\theta = \int_0^\pi \frac{\mathrm{d}\theta}{\left(x - \sqrt{x^2-1}\cos(\theta)\right)^{n+1}}$$

- Problems S3M2/3/4 formulated:
  - Problems S3M2/3/4: deduce in at least 3 distinct ways an alternative expression (a product) for the following finite sum:

$$\sum_{k=0}^{n} \cos(a + k\theta)$$

Season 3 Episode 3: One Telescope, Two Telescopes

- Problem S3M2 solved

Season 3 Episode 4: A Complex Approach

- Problem S3M3 solved

Season 3 Episode 5: Euclid Says

- Problems S3M4 solved
- Problem S3P1 formulated:
  - S3P1: find the period of oscillation of a triangular spring oscillator

Season 3 Episode 6: Triangular Spring Oscillator

- Problem S3P1 solved
- Problem S3CS1 formulated:
  - S3CS1: sort 3 integers creatively, without the popular containers and/or industrial strength algorithms

Season 3 Episode 7: 3-Integer Sort, A Discovery

- Problem S3CS1 solved

- Problem S3M5 formulated:
  - S3M5: construct a point, N, inside of a right triangle, ABC, such that the magnitudes of all the angles NAB, NBC and NCA are all equal one another

Season 3 Episode 8: Brocard Points Introduction

- Problem S3M5 solved
- Problem S3M6 formulated:
  - S3M6: invent a way to evaluate finite sums based on a popular approach that is used in combinatorial analysis to prove binomial identities

Season 3 Episode 9: Summation By Double-Counting

- Problem S3M6 solved
- Problem S3P2 formulated:
  - S3P2: find the distance covered by a bouncing material point

Season 3 Episode 10: Infinity In Physics

- Problem S3P2 solved
- Problem S3CS2 formulated:
  - S3CS2: suggest a Turing Machine-friendly algorithm for testing if a given point is located outside or inside of a given triangle

Season 3 Episode 11: A Point-in-triangle Detection Algorithm

- Problem S3CS2 solved
- Problem S3M7 formulated:
  - S3M7: find a way to integrate a binomial differential $x^m(a+bx^n)^p\,dx$

Season 3 Episode 12: Integration Of Binomial Differentials

- Problem S3M7 solved
- Problem S3M8 formulated:
  - S3M8: use the integration of binomial differentials machinery developed in the previous problem in order to evaluate the following infinite sum:

$$\sum_{n=1}^{+\infty} \frac{1}{n(n+1)(n+2)\dots(n+p)}, \quad p \geqslant 2$$

Season 3 Episode 13: Binomial Differentials Slay Infinite Sums Before Breakfast

- Problem S3M8 solved
- Problem S3P3 formulated:
  - S3P3: distinguish a sphere from an identical ball of same radius and mass

Season 3 Episode 14: Sisyphus Versus Spherical Chickens Of Uniform Density
- Problem S3P3 solved
- Problem S3CS3 formulated:
  - S3CS3: construct a single-pass algorithm for sorting **0**s and **1**s

Season 3 Episode 15: Single-pass Janus Sort
- Problem S3CS3 solved
- Problem S3M9 formulated:
  - S3M9: construct an equilateral triangle on 3 parallel straight lines

Season 3 Episode 16: An Equilateral Triangle On 3 Parallel Straight Lines Construction
- Problem S3M9 solved
- Problem S3M10 formulated:
  - S3M10: reconstruct a square given 4 points that belong to one side of that square each

Season 3 Episode 17: 4-Point Square Reconstruction
- Problem S3M10 solved
- Problem S3P4 formulated:
  - S3P4: find the normal and the tangential accelerations as functions of time of a point that moves along a parabola with its acceleration vector staying parallel to the y-axis at all times

Season 3 Episode 18: A Parabolic Motion With Acceleration Staying Parallel To The y-axis
- Problem S3P4 solved
- Problem S3CS4 formulated:
  - S3CS4: creatively swap 2 sub-arrays

Season 3 Episode 19: A Creative Swap Of Two Sub-Arrays
- Problem S3CS4 solved
- Problem S3M11 formulated:
  - S3M11: evaluate a finite trigonometric sum

$$\sum_{k=1}^{n} \frac{1}{\sin\left(2^k x\right)}$$

Season 3 Episode 20: A Finite Trigonometric Sum Evaluation
- Problem S3M11 solved
- Problem S3M12 formulated:
  - S3M12: evaluate a finite binomial sum using Physics:

$$\sum_{k=0}^{n} k \cdot \binom{n}{k}$$

Season 3 Episode 21: Physics Evaluates A Finite Binomial Sum
- Problem S3M12 solved
- Problem S3P5 formulated:
  - S3P5: recover the shape of the trajectory of a point that moves in a plane in such a way that its both tangential and normal accelerations remain constant at all times

Season 3 Episode 22: Planar Motion With Constant Tangential And Normal Accelerations
- Problem S3P5 solved
- Problem S3CS5 formulated:
  - S3CS5: explain how the Shunting Yard Algorithm works and create a Slow-Motion Player that lets a user experiment with and witness the mechanics of that algorithm

Season 3 Episode 23: The Mechanics Of The Shunting Yard Algorithm
- Problem S3CS5 solved
- Problem S3M13 formulated:
  - S3M13: evaluate the following integrals in at least 3 different creative ways:

$$C_{a,b}(x) = \int e^{ax} \cos(bx)\, dx; \qquad S_{a,b}(x) = \int e^{ax} \sin(bx)\, dx$$

Season 3 Episode 24: 3 Creative Evaluations Of Ubiquitous Integrals
- Problem S3M13 solved
- Problem S3M14 formulated: a divisibility by 7 test
  - S3M14: design a divisibility-by-7 algorithm using the Reduce-And-Conquer approach

Season 3 Episode 25: A Divisibility-By-7 Test Design Via Reduce-And-Conquer
- Problem S3M14 solved
- Problem S3P6 formulated: a kinematic descend of a rotating line segment
  - S3M14: recover the law of motion, the trajectory, the velocity, the acceleration and the curvature of the trajectory traced by one edge of a uniformly descending line segment that also rotates in the horizontal plane with a constant angular velocity

Season 3 Episode 26: A Kinematic Descend Of A Rotating Line Segment
- Problem S3P6 solved
- Problem S3CS6 formulated: describe the mechanics and then design and implement the Narayana's Next-Linear-Permutation algorithm

Season 3 Episode 27: Narayana's Next Linear Permutation Algorithm
- Problem S3CS6 solved
- Problem S3M15 formulated. Evaluate the following finite trigonometric sum:

$$\sum_{m=1}^{n} \arctan\left(\frac{2m}{m^4 + m^2 + 2}\right)$$

Season 3 Episode 28: In Like A Lion Out Like A Lamb

- Problem S3M15 solved
- Problem S3M16 formulated. Evaluate the following limit:

$$\lim_{n\to+\infty} n^{\frac{1}{n}}$$

Season 3 Episode 29: The Art Of Manipulation Of Inequalities

- Problem S3M16 solved
- Problem S3P7 formulated: deduce an equation that describes a spherical motion of a point under a constant angle with the meridians

Season 3 Episode 30: A Deduction Of An Equation Of A Loxodrome

- Problem S3P7 solved
- Problem S3CS7 formulated: construct an $O(n)$ time and $O(1)$ space complexity algorithm for detecting if a given integer can be represented as a sum of two integers from a given array sorted in some order, ascending or descending

Season 3 Episode 31: An Integer As A Sum Of Two Elements Of A Sorted Array

- Problem S3CS7 solved
- Problem S3M17 formulated: evaluate the following indefinite integral integral

$$\int \sqrt{x^2 - a^2}\, dx$$

in at least two creative ways.

Season 3 Episode 32: Hyperbolically Speaking

- Problem S3M17 solved
- Problem S3M18 formulated: evaluate the following definite integrals

$$\int_0^{\frac{\pi}{2}} \sin^m(x)\, dx \quad \text{and} \quad \int_0^{\frac{\pi}{2}} \cos^m(x)\, dx$$

for $m = 1, 2, 3, 4, 5$ and so on.

Season 3 Episode 33: Recurrence Relations Evaluate Integrals

- Problem S3M18 solved
- Problem S3M19 formulated: show that hyperbolic functions can be defined by a direct analogy with circular functions defined via origin-centered unit circle

Season 3 Episode 34: Hyperbolic Functions Defined By A Direct Analogy With Circular Functions

- Problem S3M19 solved
- Problem S3P8 formulated: recover the law of motion of a point that is moving along a straight line that sweeps the surface of a right circular cone

Season 3 Episode 35: Twice Telescopic Rotating Antenna, The Kinematics Of

- Problem S3P8 solved
- Problem S3CS8 formulated: construct an algorithm for detecting a cycle in a linked list

Season 3 Episode 36: A Vicious Circle Or Lords Of The Ring

- Problem S3CS8 solved
- Problem S3M20 formulated: show that for all real $x \neq \pm k\pi$, where $k = 0, 1, 2, 3$ and so on, it is the case that:

$$\sin(x) = x \cdot \prod_{n=1}^{+\infty} \left(1 - \frac{x^2}{n^2\pi^2}\right)$$

Season 3 Episode 37: Sine As A Product, With Applications

- Problem S3M20 solved
- Problem S3M21 formulated: evaluate the Euler-Poisson integral

$$\int_0^{+\infty} e^{-x^2}\,dx$$

using the Squeeze Theorem.

Season 3 Episode 38: What Mathematicians Do All Day Or Euler-Poisson Integral Evaluation via the Squeeze Theorem

- Problem S3M21 solved
- Problem S3P9 formulated: solve the S2E54/S2P2 problem under the assumption that there is no external input into the rotating rod-bead system, which means that the system's angular velocity is no longer constant.

Season 3 Episode 39: A Bead Sliding Off Of A Massless Rod That Rotates With A Decreasing Angular Velocity

- Problem S3P9 solved
- Problem S3CS9 formulated: find the diamond across 8 boxes with 2 cooperating people if boxes have a random orientation, the two people must agree on the algorithm ahead of time, one person is lead into the room with the boxes to do something and then escorted out of the room, while the second person, upon entering that room, has to recover the box with the diamond

Season 3 Episode 40: Computer Science And Group Theory Find A Diamond

- Problem S3CS9 solved
- Problem S3M22 formulated: solve the following linear non-homogeneous recurrence relation:

$$t_n = 6t_{n-2} - t_{n-1} - 4, \quad t_0 = 2, t_1 = 4$$

Season 3 Episode 41: How Generating Functions Can Solve Linear Non-Homogeneous Recurrence Relations

- Problem S3M22 solved
- Problem S3M23 formulated: how many chords are formed by n distinct points located on the circumference of a circle and why?

Season 3 Episode 42: Chords Of The Ring

- Problem S3M23 solved
- Problem S3P10 formulated: two points execute a coplanar motion as follows. A point R moves along a straight line with constant velocity $\overrightarrow{v_r}$, while a point C moves (in the same plane) with a velocity, $\overrightarrow{v_c}$, that is constant by magnitude but always points at the point R. Find the trajectory of the point C, the condition and the time of the intercept of R by C, the intercept time from the moment when the vectors $\overrightarrow{v_r}$ and $\overrightarrow{v_r}$ were perpendicular and the distance $|CR|$ was d_0.

Season 3 Episode 43: The Kinematics Of Wile E. Coyote and Road Runner

- Problem S3P10 solved
- Problem S3CS10 formulated: show

Season 3 Episode 44: Lorem Ipsum

- Problem S3CS10 solved
- Problem S3M24 formulated: show

Season 3 Episode 45: Lorem Ipsum

- Problem S3M24 solved
- Problem S3M25 formulated: show

Season 3 Episode 46: Lorem Ipsum

- Problem S3M25 solved
- Problem S3P11 formulated: show

Season 3 Episode 47: Lorem Ipsum

- Problem S3P11 solved
- Problem S3CS11 formulated: show

Season 3 Episode 48: Lorem Ipsum

- Problem S3CS11 solved
- Problem S3M26 formulated: show

The ProbLemma's Channel Season 3 Index

Problem Number	Formulated In	Solved In
S3M1	Season 2 Episode 55	Season 3 Episode 2
S3M2	Season 3 Episode 2	Season 3 Episode 3
S3M3	Season 3 Episode 2	Season 3 Episode 4
S3M4	Season 3 Episode 2	Season 3 Episode 5
S3P1	Season 3 Episode 5	Season 3 Episode 6
S3CS1	Season 3 Episode 6	Season 3 Episode 7
S3M5	Season 3 Episode 7	Season 3 Episode 8
S3M6	Season 3 Episode 8	Season 3 Episode 9
S3P2	Season 3 Episode 9	Season 3 Episode 10
S3CS2	Season 3 Episode 10	Season 3 Episode 11
S3M7	Season 3 Episode 11	Season 3 Episode 12
S3M8	Season 3 Episode 12	Season 3 Episode 13
S3P3	Season 3 Episode 13	Season 3 Episode 14
S3CS3	Season 3 Episode 14	Season 3 Episode 15
S3M9	Season 3 Episode 15	Season 3 Episode 16
S3M10	Season 3 Episode 16	Season 3 Episode 17
S3P4	Season 3 Episode 17	Season 3 Episode 18
S3CS4	Season 3 Episode 18	Season 3 Episode 19
S3M11	Season 3 Episode 19	Season 3 Episode 20
S3M12	Season 3 Episode 20	Season 3 Episode 21
S3P5	Season 3 Episode 21	Season 3 Episode 22
S3CS5	Season 3 Episode 22	Season 3 Episode 23
S3M13	Season 3 Episode 23	Season 3 Episode 24
S3M14	Season 3 Episode 24	Season 3 Episode 25
S3P6	Season 3 Episode 25	Season 3 Episode 26
S3CS6	Season 3 Episode 26	Season 3 Episode 27
S3M15	Season 3 Episode 27	Season 3 Episode 28
S3M16	Season 3 Episode 28	Season 3 Episode 29

S3P7	Season 3 Episode 29	Season 3 Episode 30
S3CS7	Season 3 Episode 30	Season 3 Episode 31
S3M17	Season 3 Episode 31	Season 3 Episode 32
S3M18	Season 3 Episode 32	Season 3 Episode 33
S3M19	Season 3 Episode 33	Season 3 Episode 34
S3P8	Season 3 Episode 34	Season 3 Episode 35
S3CS8	Season 3 Episode 35	Season 3 Episode 36
S3M20	Season 3 Episode 36	Season 3 Episode 37
S3M21	Season 3 Episode 37	Season 3 Episode 38
S3PS9	Season 3 Episode 38	Season 3 Episode 39
S3CS9	Season 3 Episode 39	Season 3 Episode 40
S3M22	Season 3 Episode 40	Season 3 Episode 41
S3M23	Season 3 Episode 41	Season 3 Episode 42
S3P10	Season 3 Episode 42	Season 3 Episode 43
S3CS10	Season 3 Episode 43	Season 3 Episode 44
S3M24	Season 3 Episode 44	Season 3 Episode 41
S3M44	Season 3 Episode 41	Season 3 Episode 42
S3M45	Season 3 Episode 42	Season 3 Episode 43
S3M46	Season 3 Episode 43	Season 3 Episode 44
S3M47	Season 3 Episode 44	Season 3 Episode 45
S3M48	Season 3 Episode 45	Season 3 Episode 46
S3M49	Season 3 Episode 46	Season 3 Episode 47
S3M50	Season 3 Episode 47	Season 3 Episode 48
S3M51	Season 3 Episode 48	Season 3 Episode 49
S3CS3	Season 3 Episode 49	Season 3 Episode 50
S3P1	Season 3 Episode 50	Season 3 Episode 51
S3M52	Season 3 Episode 51	Season 3 Episode 52
S3M53	Season 3 Episode 52	Season 3 Episode 53
S3P2	Season 3 Episode 53	Season 3 Episode 54
S3M54	Season 3 Episode 54	Season 3 Episode 55
S3M55	Season 3 Episode 55	Season 3 Episode 56
